Harigovind Kuruvelil

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Final-year Electrical Engineering student at Western University with hands-on experience testing and debugging electronic circuits, designing PCBs, programming microcontrollers, and working with FPGA-based systems. Comfortable using oscilloscopes, multimeters, function generators, and electronic loads to trace faults and verify hardware. Project work includes mixed-signal electronics, sensor interfaces, PWM control, embedded C/C++, soldering and PCB rework.

Education

BESc, Electrical Engineering, Western University — London, ON — Sept 2021 – Dec 2026

- **Relevant coursework:** Real-Time Embedded Systems, Microprocessors & Embedded Systems, Communication Electronics, Power Electronics, Electric Machines, Control Systems, Signal Processing, and Introduction to VLSI.

Technical Skills

- **Hardware Test & Debug:** Oscilloscopes, multimeters, function generators, electronic loads, continuity testing, signal tracing, circuit bring-up, soldering, wiring, and PCB rework.
- **PCB & Electronics:** EAGLE, Altium Designer, schematic design, PCB layout, op-amp signal conditioning, MOSFET switching, gate drivers, ADC interfaces, sensor circuits, and mixed-signal hardware.
- **Embedded & Digital Systems:** Arduino Due, Cyclone V SoC / ARM Cortex-A9, GPIO, timers, interrupts, PWM, ADC, UART, SPI, I^2C, finite-state machines, and uC/OS-II RTOS.
- **Programming:** C/C++, Python, MATLAB, Java, Bash; debugging in Visual Studio and Linux/Unix environments.
- **Engineering Tools:** Quartus, Git, Windows, Linux/Ubuntu, Visual Studio, Micro-Cap, PSCAD, and PowerWorld.

Engineering Projects

100A PWM-Controlled High-Current Power Supply — London, ON — Jan – May 2025

- Built and tested a high-current switching circuit using high-side/low-side MOSFETs, an IR2011 gate driver, op-amp signal conditioning, and Arduino control.
- Generated 20 kHz PWM, measured current and voltage signals, displayed operating values on an LCD, and logged test data to an SD card for later analysis.
- Used an oscilloscope and multimeter to trace switching faults, including shoot-through and incorrect bootstrap operation, then adjusted dead time, gate resistance, and circuit wiring while isolating the cause.
- Found a supply-routing error in the gate-driver circuit that prevented the bootstrap capacitor from charging correctly and corrected the hardware during bench testing.

Embedded Multi-Sensor Monitoring System — London, ON — Jan – May 2024

- Built an embedded monitoring system on a Cyclone V SoC with an ARM Cortex-A9, using GPIO, timers, interrupts, a PIR sensor, LCD display, and audio alerts.
- Designed and assembled a single-layer Arduino PCB in EAGLE, integrating RFID, temperature/humidity and gas sensors, servo control, and voice playback.
- Implemented SPI and I^2C communication and used a finite-state machine with interrupt-driven and polled I/O to handle sensor events and system control.
- Debugged the assembled PCB in the lab, finding incorrect pin assignments and broken ground connections and repairing the board with jumper wiring.

Real-Time ToF Tracking and Closed-Loop Control *(ECE 4460)* — London, ON — 2026

- Built a closed-loop control system on an Arduino Due running uC/OS-II, using a Time-of-Flight sensor to measure ball position and PWM to control a fan.
- Split sensing, display, setpoint, and actuator control into separate RTOS tasks and used message queues and semaphores for communication and shared hardware access.
- Increased the sensor and control update rate from 5 Hz to 20 Hz and tuned the PID controller to improve response to changes in commanded height.
- Used oscilloscope measurements and circuit testing to troubleshoot pin-mapping, wiring, and PCB faults during hardware bring-up.

Radial-Flux Outrunner BLDC Motor Prototype — London, ON

- Designed and built a 9-slot, 12-pole radial-flux BLDC motor under a 15 V / 5 A supply constraint, including winding layout, conductor selection, and permanent-magnet sizing.
- Wound and assembled the three-phase stator and used continuity checks and bench testing to troubleshoot winding connections and rotor alignment.

Jan – May 2024

Hybrid VR Spatial Audio Engine — Capstone Project

London, ON
Sept 2025 – Apr 2026

- Built and debugged a real-time C++ plugin for Unreal Engine 5 that processes changing 3D scene geometry and updates audio behaviour as the listener moves.
- Moved model inference and other heavier processing to a worker thread to avoid blocking the real-time audio path, then tested and debugged the system on a Meta Quest 3S.

Work Experience

Canada Border Services Agency (CBSA)

Mississauga, ON
May 2023 – Present

Student Border Services Officer

- Conduct inspections, review travel and immigration documents, and use government systems to process travellers and identify cases requiring additional examination.
- Maintain accurate case records while handling sensitive personal and immigration information in accordance with established procedures.
- Work independently in a busy operational environment and communicate findings to senior officers and partner agencies when a case requires additional action.
- Handle changing priorities and unusual cases while maintaining accuracy and completing required documentation.